Marvin Wu(Wu Hung Mao)
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Education

Honours Bachelors of Science (H.Bsc), Computer Science specialist. Sep. 2021 - Present
University of Toronto Mississauga (expected graduation with PEY CO-OP - August 2026) GPA: 3.45/4.0

Technical Skills

- **Languages:** Python, C, C++, Java, SQL, JavaScript, HTML/CSS, PHP
- **Frameworks & Libraries:** React, Node.js, jQuery
- **Tools & Infrastructure:** Linux, Git/GitHub/GitLab, Docker, AWS, Jenkins, Jira
- **Databases & Profiling:** PostgreSQL, SQLite, Nsight Systems, LTTng, DCGM, Valgrind
- **Other:** Agile/Scrum Methodologies, Spoken Languages (English, Mandarin)

Third/Fourth Year Courses

- **AI & Robotics:** Introduction to Machine Learning, Introduction to Artificial Intelligence, Fundamentals of Robotics
- **Systems & Security:** Operating Systems, Parallel Programming, Principles of Computer Networks, Computer Security
- **Software & Data:** Introduction to Software Engineering, Programming on the Web, Introduction to Databases
- **Theory & Algorithms:** Algorithm Design and Analysis, Computational Complexity and Computability, Principles of Programming Languages
- **Advanced Topics:** Computer Science Implementation Project, Topics in Computer Science, Computing Education

Work Experience

Physical Layer (vDU) Developer Intern May 2024 - Sep 2025
Ericsson *Ottawa, Ontario, Canada*

Project: Gphy2.0 (Feb 2025 - Present)

- Offloaded LDPC encoding onto the latest Nvidia GPU, previously running on an Intel accelerator.
- Created a new GPU branch for LDPC encoding in the existing codebase.
- Profiled latency of the GPU branch and compared Nsight System and Nsight Compute reports collected across servers with different GPUs, making adjustments to kernels to improve SM occupancy.
- Collected latency with the Intel accelerator branch calculated with the CPU cycle.
- **Environment:** Bazel(`rules_cuda`), Gerrit, Nsight System, Nsight Compute

Project: Gemini (May 2024 - Feb 2025)

- Digital twin for Radio, RF Channel, and UEs.
- Created a constellation diagram using the Ericsson Design System library and Vue3 framework.
- Wrote multi-threaded unit tests in CUDA to test a real-time system under a strict timing constraint ($\sim$ 35 µs).
- Wrote an SQLite script that drew histograms based on data collected from LTTng traces for channel quality analysis.
- **Environment:** Docker, Wireshark, Gerrit, Gitlab, Jenkins, Jfrog, Sqlite3, AWS(EC2)

Academic Papers

Course Research, University of Toronto Mississauga Apr 2026

Wu, H.-M., Bi, F., Abdelhadi, Y., Lin, W., Khan, L. Ensemble Deep Learning Approaches for AI-Altered Video Detection. Computer Science Implementation Project (CSC492).

Riku, T., Ahmed, K., **Wu, H.-M.**, Usman, R. Can Agents Debug Better Than a Software Engineer?. Course Research White Paper (CSC398).

AI & Machine Learning Projects

AI-Video Detection Ensemble | *Python, PyTorch, OpenCV, NumPy* Link | Feb 2026

- Built an ensemble composed of 4 models (EfficientNet, MesoNet, XceptionNet, AASIST) to detect general & deepfake AI-generated video.
- Fine-tuned a pre-trained EfficientNet-B1 model on datasets such as AIGVDBench and FaceForensics++, which use diffusion and face transfer methods to generate deepfake videos.
- Leveraged Transfer Learning to detect AI-generated media with high precision.
- Integrated the EfficientNet-B1 inference logic into the ensemble, utilizing MTCNN for face detection and cropping, and Albumentations for high-performance preprocessing.
- Designed an ensemble architectural diagram including preprocessing, inference, and postprocessing stages.
- Enabled parallel inference and took weighted scores to make the final REAL/FAKE prediction; communicated backend results with the frontend through HTTP requests.
- Co-authored the unpublished research manuscript "Ensemble Deep Learning Approaches for AI-Altered Video Detection," detailing the multimodal architecture, performance metrics, and ensemble strategies.
- Served as both Project Manager and Senior Developer of a 5-member team; actively participated in two weekly meetings — a Sunday sync facilitated by the mentor to review weekly progress, identify blockers, and plan upcoming tasks, and a Monday meeting to present results to the professor.

JetBrains LCA Benchmark Contribution | *Python, Docker, OpenAI API, Hugging Face API* Link | Feb 2026

- Contributed to the open-source JetBrains LCA benchmark, expanding the bug localization benchmark's capability from identifying the buggy file to automatically applying a fix.
- Used the OpenAI API to generate code patches and the Hugging Face API to programmatically download and manage large-scale repositories.
- Expanded evaluation metrics of the original benchmark by adding token usage count, iteration count, etc. for popular coding agents like Claude Code and Codex.
- Collaborated in a five-person team, aiming to support the long-term development of JetBrains' IDE-integrated agentic coder, Junie.
- Co-authored a technical white paper detailing the multi-agent architecture and evaluation metrics used to benchmark Claude Code and Codex.
- Delivered a group presentation to **40+ attendees** at the end of the project, communicating the multi-agent architecture, evaluation methodology, and benchmark findings.

Machine Learning Painting Classifier | *Python, Pandas, NumPy* Link | Feb 2026

- Used Pandas and NumPy to explore the dataset and analyze text features, identifying patterns to help the model distinguish between different paintings.
- Built a lightweight prediction script using only pure Python and NumPy because no external ML libraries are allowed, keeping the final model under 10MB for efficient deployment.
- Planned to test and tune 3 different types of models, selecting the best hyperparameters to ensure the model performs well on unseen test data rather than just memorizing the training set.

Frozen Lake Solver | *Python* Link | Feb 2025

- Implemented a reinforcement learning agent to solve the Frozen Lake environment from OpenAI Gym.
- Adopted the value iteration algorithm to find the optimal policy.

Pacman AI Agent | *Python* Link | Sep 2023 - Oct 2023

- Implemented a Pacman game agent utilizing classical search algorithms (DFS, BFS, UCS, A*) with custom heuristics for the CSC384 AI course.
- Scored 80/100 on the assignment.

Systems & High-Performance Computing Projects

Image Processing with GPU | *C, CUDA* Link | Nov 2023 - Dec 2023

- Used GPU and parallel programming to process a 10MB x 10MB PGM image in under 10 milliseconds, achieving a 70x speedup compared to sequential processing. Scored 100/100 on the assignment.
- Broke down the entire project into smaller components, demonstrating strong time management skills that led to well-formatted code and detailed documentation.
- Employed C and CUDA to parallelize execution.

Database Join with OpenMP and MPI | *C, OpenMP, MPI* Link | Oct 2023 - Nov 2023

- Performed database joins in both shared and distributed memory spaces, optimizing join speed. Achieved the merging of over ten thousand queries in under 200 milliseconds.
- Utilized C along with parallel programming concepts including OpenMP for multi-core and MPI for multi-node execution.

File System Simulation | *C, mmap, Pthreads* Link | Sep 2025 - Dec 2025

- Led a team to implement a Unix-like file system driver in C, utilizing memory mapping (mmap) to directly manipulate disk images.
- Managed disk layout structures including superblocks, inode tables, and bitmaps to track free inode and data blocks. Allocated resources as needed.
- Designed helper functions for path resolution and directory entry parsing to support nested directories and file lookups.
- Engineered logic for hard and symbolic links, managing reference counts and inode pointers to ensure data consistency.
- Designed a thread-safe synchronization scheme using fine-grained mutex locks to prevent race conditions during concurrent file operations.

Ethernet Frame and ARP Request Simulation | *C++* Link | Feb 2024

- Implemented a `NetworkInterface` in C++ that resolves IP-to-MAC addresses via ARP requests and caches replies, simulating Layer 2 switch forwarding.
- Demonstrated understanding of Ethernet frame encapsulation, ARP protocol flow, and network switch behaviour.

Web & Full-Stack Development Projects

Personal Portfolio Website | *React 18, Vite, React Router, CSS Modules, GitHub Actions* Link | Dec 2023 - Present

- Designed and built a personal portfolio at wuhungmao.github.io/academic-work with a terminal/hacker aesthetic using React 18 and Vite; features 7 pages including a typewriter home, project gallery, course reviews, publications, and an interactive timeline.
- Migrated from an original plain HTML/CSS/jQuery site (built from scratch to self-teach web development) to a component-based React architecture with CSS Modules for per-component scoped styling and React Router v6 for client-side navigation.
- Configured automated CI/CD via GitHub Actions: every push to `main` triggers a Vite build and deploys `portfolio/dist/` to GitHub Pages with no manual steps.
- Utilized vibe coding with Claude Code to rapidly iterate on features, styling, and content.

Decidophobia.com | *Django, React, JavaScript, HTML/CSS, SQL* Link | Jan 2024 - Apr 2024

- Acted as a Full Stack Developer to develop a responsive website that compares products from Amazon, Best Buy, eBay, and others, tailoring search results based on user preferences.
- Contributed to weekly sprint and non-sprint meetings, actively engaging in group discussions and proposing achievable and impactful features for implementation.
- Successfully retrieved product information from eBay and displayed it on the website.
- Monitored project progress with Jira and utilized the Django framework following the MVC model. Designed a questionnaire page using HTML, JavaScript, and CSS. Implemented back-end functionality, including product model creation and SQL queries for database storage.
- Converted a pure HTML document to React to create a more responsive website.

Microservice API | *Java, SQLite* Link | Jan 2024 - Apr 2024

– Implemented a bidirectional REST-based API, handling HTTP requests on both server and client sides. The API communicated with the database to update User and Product tables, providing HTTP responses upon completion.
– Created this project from the ground up and self-taught numerous new concepts.
– Collaborated with team members to break down the entire project into smaller, manageable tasks.
– Employed Java to interact with an SQLite database, leveraging JSON-formatted messages.

Game Collection Website | *PHP, PostgreSQL* Feb 2024

– Developed a website which incorporates different kinds of games (rock paper scissors, guessing numbers, frog puzzle).
– Developed multiple games utilizing PHP solely, incorporating concepts such as session management, HTTP requests, the Model-View-Controller (MVC) architecture, CSRF token, etc. Game information was stored in a PostgreSQL database management system.

Wordle Game | *Node.js, Express.js, jQuery* Mar 2024

– Designed a Wordle game which allows the player to guess a 5-letter word.
– Utilized the jQuery library to run JavaScript code on the front end and the Express.js framework to build a web application with Node.js and construct a RESTful API.

Other Course Projects

Boggle Game | *Java, JavaFX* Link | Oct 2022 - Dec 2022

– Led a team using Scrum methodology for Boggle game development, fostering active team communication. Successfully decomposed large group tasks, assigned them, and coordinated team efforts, resulting in a high-grade achievement.
– Created user personas and user stories, and generated detailed reports, showcasing excellent project management.
– Developed the game using Java and JavaFX, while utilizing GitHub for software development and project management.

Simon Game | *RISC-V Assembly* Link | Feb 2023 - Mar 2023

– Developed a Simon game using assembly code, designed to enhance players' short-term memory, resulting in a score of 90/100.

File Compression | *Python* Link | Jan 2022 - Feb 2022

– Used the Huffman algorithm for both file compression and decompression.